

CO2-ASSESSMENT TOOL-O2

Social Engineering Simulation

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Question 1

As a Security Analyst, perform an OSINT-based email enumeration assessment on the target domain tesla.com using theHarvester in Kali Linux. Collect publicly available email addresses, subdomains, hosts, and IP addresses associated with the organization. Categorize the discovered email addresses into likely roles such as Support, Sales, Human Resources, Administration, and Technical Teams. Analyze how the gathered information could assist an attacker during the reconnaissance phase and explain how it could be used for spear-phishing, credential harvesting, and impersonation attacks. Finally, recommend at least five countermeasures to reduce information leakage and social engineering risks, and submit a report with screenshots, commands executed, findings, risk analysis, and recommendations. (20 Marks)

Aim

To perform an OSINT-based email enumeration assessment on the target domain tesla.com using theHarvester in Kali Linux.

Objective

- Collect publicly available email addresses.
- Identify subdomains and hosts.
- Discover IP addresses.
- Categorize email addresses based on departments.
- Analyze social engineering risks.
- Recommend security countermeasures.

Tools Required

- Kali Linux
- theHarvester
- Terminal
- Internet connection

Command Used

Command 1 : theHarvester -d tesla.com -b rapiddns

Procedure

Step 1

Open Kali Linux Terminal.

Step 2

Run the command: `theHarvester -d`

tesla.com -b rapiddns

[illegible]

Command 2 : theHarvester -d tesla.com -b rapiddns -n

```
Session Actions Edit View Help
links.tesla.com:links.tesla.com.edgekey.net
livestreamapi.tesla.com:livestreamapi.tesla.com.edgekey.net
mfs-supplier.tesla.com:mfs-supplier.tesla.com.edgekey.net
mobile.tesla.com:mobile.tesla.com.edgekey.net
mta.email.tesla.com:13.111.14.190
npuhe-eng.euw1.vn.cloud.tesla.com:npuhe-eng-c-286de95ea08c5e2a.elb.eu-west-1.amazonaws.com
npuhe-eng.usw2.vn.cloud.tesla.com:npuhe-eng-a-c517f248c251dc44.elb.us-west-2.amazonaws.com
npuhe-eng.usw2.vn.cloud.tesla.com:npuhe-eng-b-979133c19331427b.elb.us-west-2.amazonaws.com
npuhe-eng.usw2.vn.cloud.tesla.com:npuhe-eng-c-ef07d5b3ed829c9b.elb.us-west-2.amazonaws.com
npute-eng.usw.vn.cloud.tesla.com:usw-eng5a-fleet-npute-prd.teslafleet.akadns.net
npute-eng.usw.vn.cloud.tesla.com:usw-eng3c-fleet-npute-prd.teslafleet.akadns.net
npute-prd.usw.vn.cloud.tesla.com:usw-prd5b-fleet-npute-prd.teslafleet.akadns.net
npute-prd.usw2.vn.cloud.tesla.com:npude-prd-a-ea25687038eac2e.elb.us-west-2.amazonaws.com
npuv-ap-prd.euw1.vn.cloud.tesla.com:npuv-prd.euw1.vn.cloud.tesla.com
npuv-eng.euw1.vn.cloud.tesla.com:npuv-eng-a-38adfa93f5795d7c.elb.eu-west-1.amazonaws.com
npuv-eng.usw.vn.cloud.tesla.com:usw-eng5d-fleet-npuv-prd.teslafleet.akadns.net
npuv-eng.usw2.vn.cloud.tesla.com:npuv-eng-a-97483e916dd01bb4.elb.us-west-2.amazonaws.com
npuv-prd.euw1.vn.cloud.tesla.com:npuv-prd-a-56bc286e6fb51aaf.elb.eu-west-1.amazonaws.com
npuv-prd.euw1.vn.cloud.tesla.com:npuv-prd-c-04d07fce0e0dc20b.elb.eu-west-1.amazonaws.com
npuv-prd.usw.vn.cloud.tesla.com:usw-prd5c-fleet-npuv-prd.teslafleet.akadns.net
npuv-prd.usw2.vn.cloud.tesla.com:npuv-prd-c-a2aa85266360facf.elb.us-west-2.amazonaws.com
origin-bolt.tesla.com:clsmdcgyp.tesla.com.akadns.net
origin-courses.tesla.com:origin-courses.tesla.com.akadns.net
origin-edr.tesla.com:clsgenp.tesla.com.akadns.net
origin-einvoicing.tesla.com:warpp-einvoicing.tesla.com.akadns.net
origin-finops.tesla.com:clsmdcFinp.tesla.com.akadns.net
origin-livestream.tesla.com:k8clsgen.tesla.com.akadns.net
origin-location-services-prd.tesla.com:clsmdcvehp.tesla.com.akadns.net
origin-static-assets-profile-settings.tesla.com:clsmdcgenp.tesla.com.akadns.net
origin-tcc-gw-stg.tesla.com:edgeservices.tesla.com.akadns.net
origin-warpp-akamai.tesla.com:199.66.9.77
partners.tesla.com:partners.tesla.com.edgekey.net
sales-prd.tesla.com:external-na-pop1.edgekey.net
shop.tesla.com:shop.tesla.com.edgekey.net
static-assets-pay.tesla.com:static-assets-pay.tesla.com-v1.edgekey.net
static-assets.tesla.com:static-assets.tesla.com.edgekey.net
stg-uc-twilio-bpm-webhook.tesla.com:external-na-pop1.edgekey.net
suppliers.tesla.com:suppliers.tesla.com.edgekey.net
toolbox.tesla.com:toolbox.tesla.com.edgekey.net
trt.tesla.com:external-na-pop1.edgekey.net
ul.engage.tesla.com:d1e601ajz6qsud.cloudfront.net
url2547.tesla.com:sendgrid.net
url5196.tesla.com:sendgrid.net
url7051.tesla.com:url7051.tesla.com.edgekey.net

[+] Starting active queries for DNSLookup.
[+] Hosts found after reverse lookup (in target domain):
```

```
Session Actions Edit View Help
url7051.tesla.com:url7051.tesla.com.edgekey.net

kali@kali: ~
$ theHarvester -d tesla.com -b rapiddns -n
Read proxies.yaml from /etc/theharvester/proxies.yaml
*****
* THE HARVESTER *
*****
* theharvester & id:1 *
* Coded by Christian Martorella *
* Edge-Security Research *
* cmartorella@edge-security.com *
*****

[+] Target: tesla.com
[+] Searching RapidDns.
[+] No IPs found.
[+] No emails found.
[+] No people found.
[+] Hosts found: 88

akamai-apigateway-automation-billing.tesla.com:akamai-apigateway-automation-billing.tesla.com.edgekey.net
akamai-apigateway-automation.tesla.com:akamai-apigateway-automation.tesla.com.edgekey.net
akamai-apigateway-captiveunderwriting.tesla.com:akamai-apigateway-captiveunderwriting.tesla.com.edgekey.net
akamai-apigateway-dev-warppmapiserver.tesla.com:akamai-apigateway-dev-warppmapiserver.tesla.com.edgekey.net
akamai-apigateway-logisticsratesapi.tesla.com:akamai-apigateway-logisticsratesapi.tesla.com.edgekey.net
akamai-apigateway-procurementpayapi.tesla.com:akamai-apigateway-procurementpayapi.tesla.com.edgekey.net
akamai-apigateway-shipmentplanningapi.tesla.com:akamai-apigateway-shipmentplanningapi.tesla.com.edgekey.net
akamai-apigateway-stg-captiveunderwriting.tesla.com:akamai-apigateway-stg-captiveunderwriting.tesla.com.edgekey.net
akamai-apigateway-stg-deliveryopsapi.tesla.com:akamai-apigateway-stg-deliveryopsapi.tesla.com.edgekey.net
akamai-apigateway-stg-finplateng-defi.tesla.com:akamai-apigateway-stg-finplateng-defi.tesla.com.edgekey.net
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akamai-apigateway-stg-inventorystxnextapi.tesla.com:akamai-apigateway-stg-inventorystxnextapi.tesla.com.edgekey.net
akamai-apigateway-stg-translationpipeline.tesla.com:akamai-apigateway-stg-translationpipeline.tesla.com.edgekey.net
akamai-apigateway-vfx.tesla.com:akamai-apigateway-vfx.tesla.com.edgekey.net
akamai-apigateway-warppapi.tesla.com:akamai-apigateway-warppapi.tesla.com.edgekey.net
auth.tesla.com:auth.tesla.com.edgekey.net
autobidder-proprd.powerhub.energy.tesla.com:energy.tesla.com.edgekey.net
autodiscover.tesla.com:autodiscover.outlook.com
```

Step 3

Collect the following information:

- Email addresses
- Subdomains • Hosts
- IP addresses

Step 4

Categorize the email addresses into:

- Support Team
- Sales Team
- Human Resources
- Administration Team
- Technical Team

Risk Analysis

The collected information can be used for:

- Phishing attacks
- Credential theft
- Employee impersonation
- Password attacks
- Social engineering attacks

Countermeasures

1. Enable multi-factor authentication.
2. Hide sensitive employee emails.
3. Conduct security awareness training.
4. Use email filtering.
5. Monitor suspicious activities.

Conclusion

OSINT tools can expose sensitive information that attackers may exploit. Organizations must protect their public data and train employees against social engineering attacks.

Question 2

As a cybersecurity awareness consultant, conduct a phishing awareness simulation using GoPhish and dummy employee accounts provided in the lab. Create an email template, launch a campaign, monitor user interactions, and generate a report showing delivery rate, email open rate, and click rate. Based on the results, provide recommendations to improve user awareness against phishing attacks. (20 Marks) Aim

To conduct a phishing awareness simulation using GoPhish and analyze user interactions.

Objective

- Create a phishing email template.
- Launch a phishing campaign.
- Monitor employee responses.
- Measure delivery, open, and click rates.
- Improve employee awareness.

Tools Required

- Kali Linux
- GoPhish
- Web browser

Command 1 : sudo apt update[illegible]


```
Command 2 : wget https://github.com/gophish/gophish/releases/latest/download/gophish-v0.12.1-linux-64bit.zip
```

Command 3 : `unzip gophish-v0.12.1-linux-64bit.zip -d gophish`

Command 4 : cd gophish

Command 5: `chmod +x gophish` **Command 5:** `sudo ./gophish`

[illegible]

Procedure

Step 1

Open the GoPhish dashboard.

Step 2

Create employee groups.

Step 3

Create an email template.

Step 4

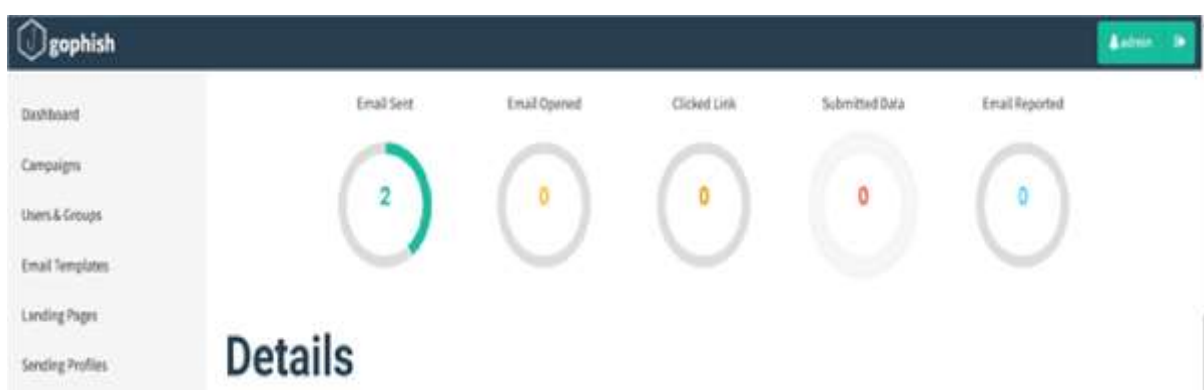
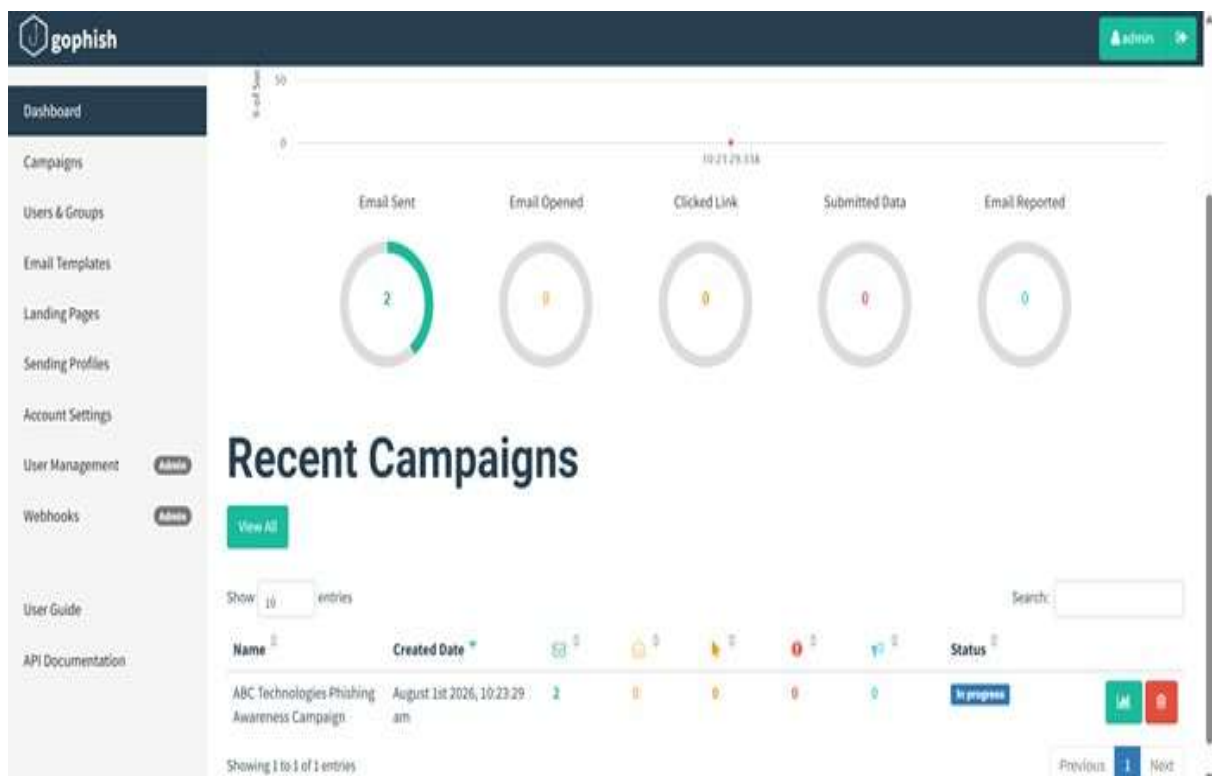
Create a landing page.

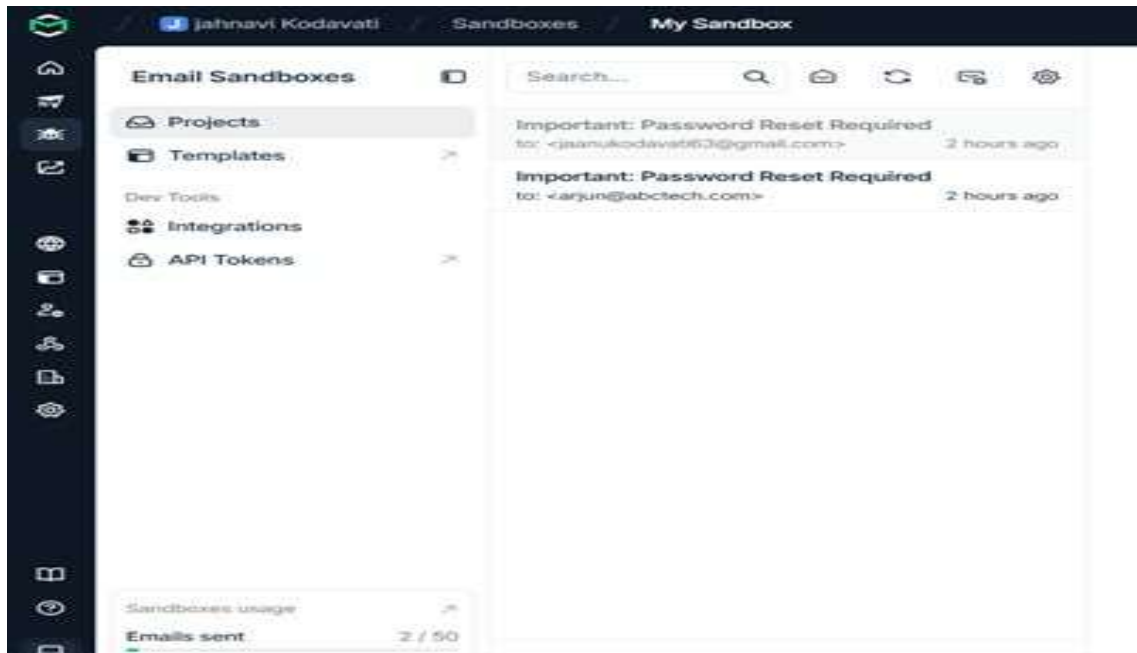
Step 5

Launch the campaign.

Step 6

Monitor the results.





Recommendations

- Verify sender emails.
- Avoid unknown links.
- Use strong passwords.
- Enable multi-factor authentication.
- Report suspicious emails.

Conclusion

Phishing simulations help organizations improve employee awareness and reduce cyber risks.

Question 3

Using Nmap Aggressive Scan (-A) on scanme.nmap.org, gather detailed information about the target, including operating system details, open ports, and service versions. Assume you are an attacker and describe how the collected information could be leveraged to conduct a social engineering attack against the IT support team. Prepare a report explaining the attack methodology and defensive recommendations. (10 Marks)

Aim

To perform an aggressive scan on the target scanme.nmap.org using Nmap.

Objective

- Identify open ports.

- Detect services.
- Find operating system details.
- Analyze security risks.
- Recommend defensive measures.

Tool Required

- Kali Linux
- Nmap

Command Used

Command 1 : nmap -A scanme.nmap.org

Procedure

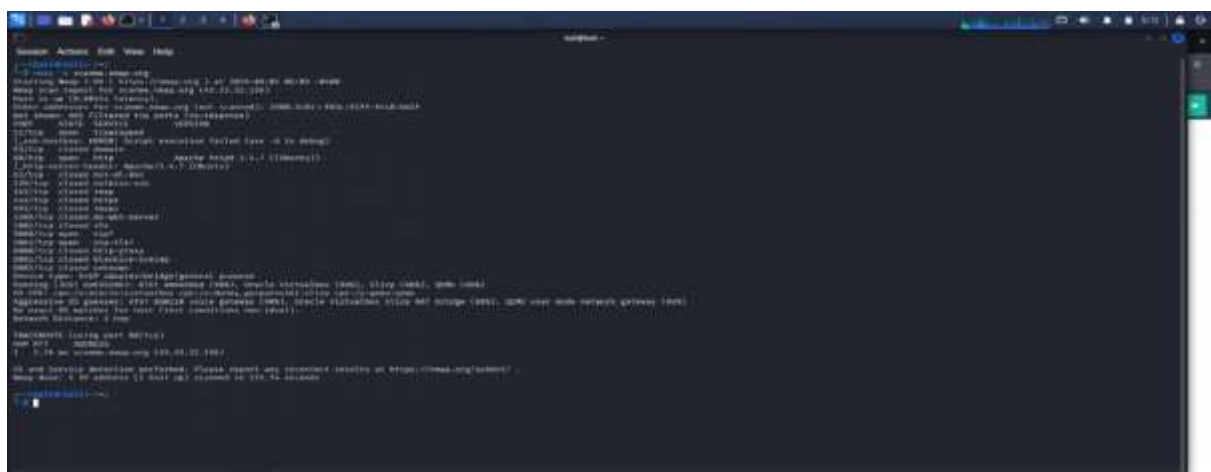
Step 1

Open Kali Linux.

Step 2

Run the command:

```
nmap -A scanme.nmap.org
```



Step 3

Collect information such as:

- Open ports
- Service versions

- Operating system details

Social Engineering Scenario An

attacker can:

- Impersonate the IT team.
- Send fake emails.
- Ask employees for credentials.
- Exploit exposed services.

Defensive Recommendations 1.

Close unnecessary ports.

2. Use firewalls.
3. Update software regularly.
4. Train employees.
5. Use strong passwords.

Conclusion

Nmap helps security professionals identify vulnerabilities and improve the security of computer networks.